MSG No. 142.567 $I4_1/ac'd'$ [Type III, tetragonal]

Table 1: Wyckoff site: `8a`, site symmetry: `-4..`

No.	position	mapping
1	$\left[0, \frac{1}{4}, \frac{3}{8}\right]$	`[1,4,6,7]`
2	$\left[0, \frac{3}{4}, \frac{5}{8}\right]$	`[2,3,5,8]`
3	$\left[\frac{1}{2}, \frac{1}{4}, \frac{5}{8}\right]$	`[9,10,15,16]`
4	$\left[0, \frac{1}{4}, \frac{7}{8}\right]$	`[11,12,13,14]`
5	$\left[\frac{1}{2}, \frac{3}{4}, \frac{7}{8}\right]$	`[17,20,22,23]`
6	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{8}\right]$	`[18,19,21,24]`
7	$\left[0, \frac{3}{4}, \frac{1}{8}\right]$	`[25,26,31,32]`
8	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{8}\right]$	`[27,28,29,30]`

Table 2: Wyckoff site: `8b`, site symmetry: `2.2'2'`

No.	position	mapping
1	$\left[0, \frac{1}{4}, \frac{1}{8}\right]$	`[1,4,11,12]`
2	$\left[0, \frac{3}{4}, \frac{3}{8}\right]$	`[2,3,25,26]`
3	$\left[0, \frac{3}{4}, \frac{7}{8}\right]$	`[5,8,31,32]`
4	$\left[0, \frac{1}{4}, \frac{5}{8}\right]$	`[6,7,13,14]`
5	$\left[\frac{1}{2}, \frac{1}{4}, \frac{7}{8}\right]$	`[9,10,18,19]`
6	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{8}\right]$	`[15,16,21,24]`
7	$\left[\frac{1}{2}, \frac{3}{4}, \frac{5}{8}\right]$	`[17,20,27,28]`
8	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{8}\right]$	`[22,23,29,30]`

Table 3: Wyckoff site: `16c`, site symmetry: `-1`

No.	position	mapping
1	$[0, 0, 0]$	`[1,5]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[2,22]`
3	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[3,23]`
4	$\left[0, \frac{1}{2}, 0\right]$	`[4,8]`
5	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[6,18]`
6	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[7,19]`
7	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,29]`
8	$\left[\frac{1}{2}, 0, 0\right]$	`[10,30]`
9	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[11,15]`
10	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[12,16]`
11	$\left[0, 0, \frac{1}{2}\right]$	`[13,25]`
12	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[14,26]`
13	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[17,21]`
14	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[20,24]`
15	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[27,31]`
16	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[28,32]`

Table 4: Wyckoff site: `16d`, site symmetry: `2..`

No.	position	mapping
1	$\left[0,\ \frac{1}{4},\ z\right]$	`[1,4]`
2	$\left[0,\ \frac{3}{4},\ z+\frac{1}{4}\right]$	`[2,3]`
3	$\left[0,\ \frac{3}{4},\ -z\right]$	`[5,8]`
4	$\left[0,\ \frac{1}{4},\ \frac{3}{4}-z\right]$	`[6,7]`
5	$\left[\frac{1}{2},\ \frac{1}{4},\ -z\right]$	`[9,10]`
6	$\left[0,\ \frac{1}{4},\ \frac{1}{4}-z\right]$	`[11,12]`
7	$\left[0,\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[13,14]`
8	$\left[\frac{1}{2},\ \frac{1}{4},\ z+\frac{1}{4}\right]$	`[15,16]`
9	$\left[\frac{1}{2},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[17,20]`
10	$\left[\frac{1}{2},\ \frac{1}{4},\ z+\frac{3}{4}\right]$	`[18,19]`
11	$\left[\frac{1}{2},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	`[21,24]`
12	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{1}{4}-z\right]$	`[22,23]`
13	$\left[0,\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[25,26]`
14	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{3}{4}-z\right]$	`[27,28]`
15	$\left[\frac{1}{2},\ \frac{3}{4},\ z\right]$	`[29,30]`
16	$\left[0,\ \frac{3}{4},\ z+\frac{3}{4}\right]$	`[31,32]`

Table 5: Wyckoff site: `16e`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,25]`
2	$\left[\frac{1}{4},\ x+\frac{3}{4},\ \frac{1}{2}\right]$	`[2,27]`
3	$\left[\frac{3}{4},\ \frac{3}{4}-x,\ \frac{1}{2}\right]$	`[3,28]`
4	$\left[-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[4,26]`
5	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[5,13]`
6	$\left[\frac{3}{4},\ \frac{1}{4}-x,\ \frac{1}{2}\right]$	`[6,15]`
7	$\left[\frac{1}{4},\ x+\frac{1}{4},\ \frac{1}{2}\right]$	`[7,16]`
8	$\left[x,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[8,14]`
9	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[9,17]`
10	$\left[\frac{1}{2}-x,\ 0,\ \frac{3}{4}\right]$	`[10,20]`
11	$\left[\frac{3}{4},\ x+\frac{1}{4},\ 0\right]$	`[11,18]`
12	$\left[\frac{1}{4},\ \frac{1}{4}-x,\ 0\right]$	`[12,19]`
13	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[21,29]`
14	$\left[\frac{1}{4},\ \frac{3}{4}-x,\ 0\right]$	`[22,31]`
15	$\left[\frac{3}{4},\ x+\frac{3}{4},\ 0\right]$	`[23,32]`
16	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[24,30]`

Table 6: Wyckoff site: `16f`, site symmetry: `..2'`

No.	position	mapping
1	$\left[x,\ x+\frac{1}{4},\ \frac{1}{8}\right]$	`[1,11]`
2	$\left[-x,\ x+\frac{3}{4},\ \frac{3}{8}\right]$	`[2,26]`
3	$\left[x,\ \frac{3}{4}-x,\ \frac{3}{8}\right]$	`[3,25]`
4	$\left[-x,\ \frac{1}{4}-x,\ \frac{1}{8}\right]$	`[4,12]`

continued ...

Table 6

No.	position	mapping
5	$\left[-x,\ \frac{3}{4}-x,\ \frac{7}{8}\right]$	[5,31]
6	$\left[x,\ \frac{1}{4}-x,\ \frac{5}{8}\right]$	[6,14]
7	$\left[-x,\ x+\frac{1}{4},\ \frac{5}{8}\right]$	[7,13]
8	$\left[x,\ x+\frac{3}{4},\ \frac{7}{8}\right]$	[8,32]
9	$\left[x+\frac{1}{2},\ \frac{1}{4}-x,\ \frac{7}{8}\right]$	[9,19]
10	$\left[\frac{1}{2}-x,\ x+\frac{1}{4},\ \frac{7}{8}\right]$	[10,18]
11	$\left[\frac{1}{2}-x,\ \frac{1}{4}-x,\ \frac{3}{8}\right]$	[15,21]
12	$\left[x+\frac{1}{2},\ x+\frac{1}{4},\ \frac{3}{8}\right]$	[16,24]
13	$\left[x+\frac{1}{2},\ x+\frac{3}{4},\ \frac{5}{8}\right]$	[17,27]
14	$\left[\frac{1}{2}-x,\ \frac{3}{4}-x,\ \frac{5}{8}\right]$	[20,28]
15	$\left[x+\frac{1}{2},\ \frac{3}{4}-x,\ \frac{1}{8}\right]$	[22,30]
16	$\left[\frac{1}{2}-x,\ x+\frac{3}{4},\ \frac{1}{8}\right]$	[23,29]

Table 7: Wyckoff site: 32g, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[\frac{1}{4}-y,\ x+\frac{3}{4},\ z+\frac{1}{4}\right]$	[2]
3	$\left[y+\frac{3}{4},\ \frac{3}{4}-x,\ z+\frac{1}{4}\right]$	[3]
4	$\left[-x,\ \frac{1}{2}-y,\ z\right]$	[4]
5	$[-x,\ -y,\ -z]$	[5]
6	$\left[y+\frac{3}{4},\ \frac{1}{4}-x,\ \frac{3}{4}-z\right]$	[6]
7	$\left[\frac{1}{4}-y,\ x+\frac{1}{4},\ \frac{3}{4}-z\right]$	[7]
8	$\left[x,\ y+\frac{1}{2},\ -z\right]$	[8]
9	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[9]
10	$\left[\frac{1}{2}-x,\ y,\ -z\right]$	[10]
11	$\left[y+\frac{3}{4},\ x+\frac{1}{4},\ \frac{1}{4}-z\right]$	[11]
12	$\left[\frac{1}{4}-y,\ \frac{1}{4}-x,\ \frac{1}{4}-z\right]$	[12]
13	$\left[-x,\ y,\ z+\frac{1}{2}\right]$	[13]
14	$\left[x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[14]
15	$\left[\frac{3}{4}-y,\ \frac{1}{4}-x,\ z+\frac{1}{4}\right]$	[15]
16	$\left[y+\frac{1}{4},\ x+\frac{1}{4},\ z+\frac{1}{4}\right]$	[16]
17	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[17]
18	$\left[\frac{3}{4}-y,\ x+\frac{1}{4},\ z+\frac{3}{4}\right]$	[18]
19	$\left[y+\frac{1}{4},\ \frac{1}{4}-x,\ z+\frac{3}{4}\right]$	[19]
20	$\left[\frac{1}{2}-x,\ -y,\ z+\frac{1}{2}\right]$	[20]
21	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[21]
22	$\left[y+\frac{1}{4},\ \frac{3}{4}-x,\ \frac{1}{4}-z\right]$	[22]
23	$\left[\frac{3}{4}-y,\ x+\frac{3}{4},\ \frac{1}{4}-z\right]$	[23]
24	$\left[x+\frac{1}{2},\ y,\ \frac{1}{2}-z\right]$	[24]
25	$\left[x,\ -y,\ \frac{1}{2}-z\right]$	[25]
26	$\left[-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[26]
27	$\left[y+\frac{1}{4},\ x+\frac{3}{4},\ \frac{3}{4}-z\right]$	[27]
28	$\left[\frac{3}{4}-y,\ \frac{3}{4}-x,\ \frac{3}{4}-z\right]$	[28]
29	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z\right]$	[29]

continued ...

Table 7

No.	position	mapping
30	$\left[x+\frac{1}{2},\,-y,\,z\right]$	[30]
31	$\left[\frac{1}{4}-y,\,\frac{3}{4}-x,\,z+\frac{3}{4}\right]$	[31]
32	$\left[y+\frac{3}{4},\,x+\frac{3}{4},\,z+\frac{3}{4}\right]$	[32]